

Diagnosis, Treatment, and Prognosis of Cerebral Venous Thrombosis

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Abstract

Cerebral venous thrombosis (CVT) is an uncommon but potentially severe cerebrovascular disorder that predominantly affects young adults and women. Advances in neuroimaging have improved diagnostic accuracy and have enabled better characterization of clinically relevant phenotypes, including edema patterns, venous infarction, and hemorrhagic transformation, which may influence prognosis and therapeutic strategies. Anticoagulation remains the cornerstone of treatment across disease severities. Emerging evidence supports the use of direct oral anticoagulants as safe and effective alternatives to vitamin K antagonists in selected patients. The role of endovascular therapy remains limited to carefully selected cases with clinical deterioration, despite optimal medical therapy, while decompressive surgery may be lifesaving in severe CVT with large space-occupying lesions and risk of cerebral herniation. Improved recognition of systemic and local prothrombotic risk factors has refined secondary prevention strategies and highlights the need for individualized management. Long-term outcomes extend beyond functional recovery, with many patients experiencing persistent cognitive symptoms, fatigue, and reduced quality of life. Despite recent progress, important gaps remain, including optimal management of severe CVT and improved tools for early prognostic stratification integrating clinical, imaging, and biomarker data. Future research should focus on personalized management approaches and strategies to reduce global disparities in CVT care.

Introduction

Cerebral venous thrombosis (CVT) is a rare yet potentially life-threatening cerebrovascular disorder, primarily affecting young adults and women. The estimated annual incidence ranges from 1.3 to 1.6 cases per 100,000 person-years, although recent studies suggest that incidence may be increasing because of improved neuroimaging availability and greater clinical awareness.¹⁻³ The epidemiology of CVT varies across world regions. In low- and middle-income countries, infectious etiologies, dehydration, and delayed access to specialized care may contribute substantially to disease burden. By contrast, the etiologic spectrum in high-income settings is more frequently dominated by malignancy, autoimmune disorders, and other prothrombotic conditions. These regional differences likely reflect disparities in health care access, diagnostic resources, and risk factors distribution.¹⁻⁴

CVT is associated with a broad spectrum of acquired and inherited prothrombotic conditions. Common acquired risk factors include pregnancy, puerperium, oral contraceptive use, malignancy, infections, autoimmune disorders, trauma, and dehydration, whereas inherited thrombophilias, such as factor V Leiden mutation, prothrombin gene mutation, and deficiencies of protein C, protein S, or antithrombin, can also contribute to susceptibility to the disease, particularly in selected patients.^{1,2} Malignancy has emerged as an increasingly recognized cause of CVT, particularly in patients with solid tumors and hematologic cancers, likely reflecting cancer-associated hypercoagulability, systemic inflammation, and treatment-related factors.⁵

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Glossary

AHA/ASA = American Heart Association/American Stroke Association; **ASM** = antiseizure medication; **CTV** = CT venography; **CVT** = cerebral venous thrombosis; **DC** = decompressive craniectomy; **DOAC** = direct oral anticoagulant; **EVT** = endovascular therapy; **IHS** = intracranial hypertension syndrome; **LMWH** = low-molecular-weight heparin; **mRS** = modified Rankin Scale; **MRV** = magnetic resonance venography; **UFH** = unfractionated heparin; **VITT** = vaccine-induced immune thrombotic thrombocytopenia; **VKA** = vitamin K antagonist.

Trauma, especially head injury involving skull fractures or direct venous sinus damage, represents another important but often underrecognized etiology.⁶ These diverse etiologies contribute substantially to the clinical heterogeneity of CVT and may influence both treatment decisions and prognosis.

Unlike arterial strokes, CVT results from thrombosis of the cerebral venous system, leading to venous congestion, blood-brain barrier disruption, and potential intracranial hemorrhage.^{1,2} The clinical presentation of CVT is heterogeneous, ranging from isolated headache to focal deficits, seizures, encephalopathy, or features of intracranial hypertension syndrome (IHS), which contributes to diagnostic delay and variability in early management.^{1,2} Advances in neuroimaging and an improved understanding of CVT pathophysiology have enhanced early detection and treatment strategies.³

This narrative review aims to critically appraise recent advances in the diagnosis, treatment, and prognostic stratification of CVT, with the goal of translating emerging evidence into a practical, decision-oriented framework for adult CVT management.

Diagnosis of CVT

CVT most commonly presents with a subacute onset (>48 hours to 1 month), although acute and chronic presentations may also occur.¹⁻³ Its clinical presentation is highly heterogeneous and frequently nonspecific, which contributes to diagnostic delay in routine practice. Observational studies report a median diagnostic delay of 4–7 days, and up to 20% of patients are diagnosed more than 1 week after symptom onset.⁷ It is important that delayed recognition is associated with preventable complications, including hemorrhagic transformation, and long-term neurologic disability, underscoring the need for early clinical suspicion and prompt venous neuroimaging.^{7,8}

Headache is the leading presenting symptom of CVT, reported in approximately 80%–90% of cases, and is commonly accompanied by focal neurologic deficits, seizures, or altered mental status.¹ However, because headache may be the sole initial manifestation, it is frequently misattributed to primary headache disorders or other secondary causes, contributing to diagnostic delay. Clinical suspicion should be heightened when headache is progressive, persistent, of thunderclap onset, or associated with papilledema or changes in consciousness.⁹

Beyond headache, CVT manifests through recognizable clinical syndromes that reflect the extent and location of venous involvement. These include IHS, focal neurologic deficits (hemiparesis or aphasia, depending on the location of the thrombus), seizures (occurring in up to 40% of cases, often with a focal onset), and encephalopathy ranging from confusion and lethargy to coma, particularly in severe cases or when the deep venous system is involved. Recognizing these syndromic patterns, rather than isolated symptoms, is critical to prompt neuroimaging and early diagnosis.^{1,2,9,10}

Definitive diagnosis of CVT requires demonstration of venous thrombosis or absence of venous opacification on vascular imaging. MRI combined with magnetic resonance venography (MRV) remains the preferred imaging modality because it enables visualization of intraluminal thrombus and flow abnormalities. The addition of susceptibility-weighted imaging or T2* sequences improves detection of acute thrombus and isolated cortical vein thrombosis, which may be missed on conventional sequences.¹¹

CT venography (CTV) represents a widely available and reliable alternative, particularly in the acute setting, with reported sensitivity of 85%–100% and specificity of 82%–95% for venous sinus thrombosis.¹² Although noncontrast CT may provide supportive clues—such as the hyperdense sinus or cord sign, and the empty delta sign on contrast-enhanced studies—, these findings are indirect and insufficient for diagnosis in isolation.

D-dimer may be elevated in CVT, but diagnostic performance is inconsistent. Sensitivity decreases in patients with symptom duration longer than 7 days, isolated headache or IHS, or limited clot burden.¹³ Current guidelines do not recommend D-dimer as a standalone diagnostic test and emphasize that imaging confirmation is mandatory when CVT is clinically suspected.¹⁴

Causes and Associated Conditions

COVID-19 and COVID-19 Vaccination

Severe acute respiratory syndrome coronavirus 2 infection is associated with an increased risk of venous and cerebral thrombosis through endothelial injury, systemic inflammation, and hypercoagulability.¹⁵ Large population-based studies consistently demonstrate that the risk of CVT is substantially higher after COVID-19 infection than after

vaccination, with reported rates of approximately 30–40 cases per million after infection compared with fewer than 1 case per million after messenger RNA vaccination.^{16,17} Rare cases of CVT have been reported after adenoviral vector vaccines (ChAdOx1 nCoV-19 and Ad26.COV2.S) in the context of vaccine-induced immune thrombotic thrombocytopenia (VITT), an immune-mediated syndrome characterized by thrombosis, thrombocytopenia, elevated D-dimer levels, and anti-platelet factor 4 antibodies. VITT typically occurs within 5–30 days after vaccination, and CVT is among its most frequent and severe thrombotic manifestations.¹⁷ Although exceedingly uncommon, early recognition is critical because management differs from conventional CVT and includes prompt nonheparin anticoagulation and IV immunoglobulin therapy.^{15–17}

Occult Cancer

Cancer is identified in approximately 5%–10% of patients at the time of CVT diagnosis.^{2,4} In addition, population-based studies suggest that subsequent cancer diagnoses may occur after CVT, with a cumulative incidence of 5.9% at 10 years.⁵ However, current evidence remains insufficient to support routine extensive cancer screening in all patients with CVT. Occult malignancy should be considered in patients with apparently unprovoked CVT, particularly in older adults (>50 years) and men. A targeted diagnostic approach based on these clinical features, basic laboratory evaluation, and age-appropriate cancer screening seems most appropriate.¹⁸

Head Trauma

Head trauma is an underrecognized risk factor for CVT. Reported prevalence is approximately 4% among patients with traumatic brain injury and may reach 20%–40% in the presence of skull fractures involving adjacent dural venous sinuses. Proposed mechanisms include endothelial injury, local inflammation, venous sinus compression, and trauma-associated hypercoagulability. However, available evidence remains largely retrospective, and the true incidence and clinical significance of trauma-associated CVT are not yet fully defined.^{6,19}

Acute Treatment of CVT

Therapeutic anticoagulation is the cornerstone of acute CVT treatment and should be initiated even in the presence of intracerebral hemorrhage.^{1,2,14} Current guidelines and expert consensus favor low-molecular-weight heparin (LMWH) over unfractionated heparin (UFH) as first-line therapy, based on its more predictable pharmacokinetics and lower risk of complications. UFH is reserved for selected situations requiring rapid reversibility, anticipated invasive procedures, severe renal impairment, or close dose titration in critically ill patients.^{1,14,18}

Vitamin K antagonists (VKAs) have traditionally been used for oral anticoagulation after initial heparin therapy, targeting an International Normalized Ratio of 2.0–3.0. However,

direct oral anticoagulants (DOACs) have emerged as a reasonable and increasingly supported alternative in many patients.^{20–24}

Randomized and observational evidence supporting DOAC use in CVT is summarized in Table 1. The RESPECT-CVT trial randomized clinically stable patients, after 5–15 days of heparin, to dabigatran or warfarin for 24 weeks, demonstrating very low rates of recurrent venous thromboembolism, similar major bleeding, and comparable recanalization (~60%–67%).²⁰ The SECRET feasibility randomized clinical trial suggested that rivaroxaban was associated with low rates of recurrent events, although bleeding was numerically higher than standard-of-care anticoagulation (warfarin, target international normalized ratio, 2.0–3.0, or LMWH); event numbers, however, were small.²¹

Observational data further support DOAC use. In the large ACTION-CVT multicenter cohort, DOACs were associated with similar rates of recurrent thrombosis and lower rates of major bleeding compared with warfarin.²² More recently, the CHOICE-CVT randomized trial further supported the safety and effectiveness of DOACs compared with warfarin in clinically stable CVT patients, showing comparable thrombotic and hemorrhagic outcomes.²³ These findings were reinforced by the prospective DOAC-CVT study (619 patients), which showed noninferior effectiveness and safety of DOACs vs VKAs in routine clinical practice.²⁴

Contemporary guidelines reflect this evolving evidence. The 2024 American Heart Association/American Stroke Association (AHA/ASA) scientific statement endorses early therapeutic anticoagulation, preferably with LMWH in the acute phase, and acknowledges the growing evidence supporting DOACs for oral therapy.¹⁸ Similarly, the 2024 German S2k guidelines recommend LMWH acutely and support DOAC use during the oral phase in appropriate patients.²⁵ By contrast, earlier 2017 ESO guidelines do not recommend DOACs, particularly during the acute phase, because they were developed before the publication of the major randomized and observational studies supporting their use in CVT.¹⁴

From a practical standpoint, current evidence suggests that DOACs represent a safe and effective alternative to VKAs in clinically stable patients with CVT and no contraindications. In our view, the main determinants of anticoagulant selection are not efficacy differences—because available data show comparable outcomes—but rather patient profile, bleeding risk, renal function, drug interactions, and anticipated adherence. Nevertheless, the acute phase requires careful clinical judgment and close monitoring, particularly in patients with extensive clot burden, fluctuating neurologic status, or high-risk thrombophilic conditions such as antiphospholipid syndrome, in whom VKAs remain the preferred oral anticoagulant.^{26,27} By contrast, available evidence does not support a similar concern in other prothrombotic conditions, including malignancy, where observational CVT

Table 1 Comparative Studies of DOACs and VKAs in CVT

Characteristics	RESPECT-CVT ²⁰	ACTION-CVT ²²	SECRET ²¹	CHOICE-CVT ²³	DOAC-CVT ²⁴
Year of publication	2019	2022	2023	2024	2025
Study design	Randomized, open-label, exploratory trial	Multicenter retrospective cohort	Randomized pilot trial	Randomized, open-label	Multicenter prospective registry
Sample size	120	845	53	89	619
Age	45	46	48	Predominantly <40	42
Female, %	55	63–67	66	49	63–64
Baseline severity (selected markers)	Mostly mild-moderate; CVT-Risk Score >3 in ~15% of patients	Not reported	NIHSS >5 in 15.4% vs 7.4%	NIHSS 0 in 66%	Majority GCS 15
Intracranial hemorrhage	30 vs 31.4%	37.8% vs 36.4%	52% vs 30.8%	11.4% vs 11.1%	35% vs 37%
Anticoagulation treatment	Dabigatran 50% vs VKA 50%	VKA: 438 DOAC: 279 Apixaban 66% Rivaroxaban 18.2% Dabigatran 13.5% Others <2%	Rivaroxaban 26 vs 27 VKA	Dabigatran vs warfarin	VKA: 218 DOAC: 401 Dabigatran 75% Apixaban 16% Rivaroxaban 15% Edoxaban 1%
Time to switch, d	5–7	Variable	4 vs 3	10–15	6 in both groups
Primary outcome	Composite of major bleeding or recurrent VTE	Recurrent VTE or CVT	Safety composite: all-cause mortality, sICH, and major extracranial hemorrhage	Composite of major bleeding or recurrent VTE	Recurrent VTE or major bleeding
Secondary outcome	Safety outcomes: Major bleeding event, clinically relevant nonmajor bleeding, any bleeding	Recanalization, major hemorrhage extracranial, ICHs, mortality	All-cause mortality, sICH, and major extracranial hemorrhage. Clinically relevant nonmajor bleeding, VTE recurrence	Recanalization, death, clinically relevant nonmajor bleeding	Mortality, major bleeding, mRS
Results (primary outcome)	Similar between groups	No difference in recurrence; lower major bleeding with DOAC	No significant difference	Comparable efficacy and safety vs warfarin	No significant difference in composite outcome
Key message	Demonstrated comparable safety vs warfarin in an exploratory RCT	Real-world data suggesting lower major bleeding with DOACs	Pilot RCT supporting feasibility of DOAC use in CVT	Dabigatran showed comparable efficacy/safety vs warfarin with less bleeding in stable CVT	Largest prospective registry supporting safety of DOACs

Abbreviations: CVT = cerebral venous thrombosis; DOAC = direct oral anticoagulant; GCS = Glasgow Coma Scale; ICH = intracranial hemorrhage; mRS = modified Rankin Scale; NIHSS = NIH Stroke Scale; RCT = randomized controlled trial; sICH = symptomatic intracranial hemorrhage; VKA = vitamin K antagonist; VTE = venous thromboembolism.

data have shown comparable effectiveness and safety of DOACs in appropriately selected patients.²⁸

Current guidelines recommend anticoagulation for 3–6 months in provoked CVT and 6–12 months in unprovoked cases, whereas indefinite treatment may be considered in patients with recurrent thrombosis or severe thrombophilia. The 2024 AHA/ASA scientific statement and the updated German S2k guideline support individualized treatment duration based on provoking factors, thrombophilia profile, recanalization status, and clinical course.^{18,25}

Recent data from the late-breaking EXCOA-CVT trial presented at ESOC 2026 suggest that extended anticoagulation

(12 months) may not result in a clear reduction in recurrent venous thromboembolic events compared with shorter treatment duration (3–6 months), although peer-reviewed results are awaited.²⁹

Pregnancy-Associated CVT

Pregnancy and puerperium represent major sex-specific risk factors for CVT, particularly during the third trimester and the first weeks postpartum. Management generally relies on therapeutic LMWH, which remains the anticoagulation of choice during pregnancy because of its safety profile and lack of placental transfer. VKA and DOACs are generally avoided during pregnancy because of teratogenicity and insufficient safety data, respectively.¹⁸

Future pregnancies are not contraindicated in most women with prior CVT, although recurrence risk should be individually assessed according to underlying thrombophilia, provoking factors, and obstetric history. Current recommendations generally support prophylactic LMWH during subsequent pregnancies and the postpartum period in women considered at increased thrombotic risk. Multidisciplinary management involving neurology, hematology, and maternal-fetal medicine specialists is essential to optimize maternal and fetal outcomes.^{25,30}

Seizure Management in CVT and Risk of Poststroke Epilepsy

Seizures occur in approximately 20%–40% of patients during the acute phase, and late unprovoked seizures (post-CVT epilepsy) develop in >10% during follow-up. Risk is higher in patients with early seizures, impaired consciousness or focal deficits at presentation, hemorrhagic parenchymal lesions, and superior sagittal sinus involvement.^{31–33} Routine prophylactic antiseizure medication (ASM) is not recommended in patients without clinical seizures. In patients with acute symptomatic seizures, ASM duration should be individualized based on clinical presentation, neuroimaging findings, and the estimated risk of post-CVT epilepsy because no randomized trials have established the optimal treatment duration.^{31–33}

A newly developed and externally validated score, DIAS-3, estimates 1- and 3-year epilepsy risk after CVT using 6 acute-phase variables: decompressive craniectomy (DC), intracerebral hemorrhage, age, acute seizure (excluding status epilepticus), status epilepticus, and subdural hematoma.³³ Across the derivation and 2 validation cohorts ($n \approx 2,200$), discrimination was good (C-statistics ~ 0.74 – 0.80), with predicted 1-year risk ranging from $\sim 7\%$ to 68% . DIAS-3 can inform follow-up intensity and decisions on ASM continuation vs tapering.

Although other prediction models, such as SeLECT,³⁴ IsCHEMiA,³⁵ and CAVE,³⁶ are available for estimating post-stroke epilepsy risk, these tools were developed in patients with arterial ischemic stroke or intracerebral hemorrhage and have not been validated in CVT. Therefore, DIAS-3 currently represents the most relevant risk stratification tool specifically developed for patients with CVT.³³

Management of Severe CVT

The Role of Endovascular Therapy

Observational studies have reported high recanalization rates with various endovascular therapy (EVT) techniques, including intrasinus thrombolysis and mechanical thrombectomy.³⁷ However, the TO-ACT randomized trial failed to demonstrate a clinical benefit of EVT over standard anticoagulation, with no significant differences in functional outcome, mortality, or symptomatic intracranial hemorrhage.³⁸ To date, no specific patient subgroup has been

consistently shown to benefit from EVT, and optimal selection criteria and procedural strategy remain undefined.

The discrepancy between high angiographic recanalization rates and the absence of demonstrated functional benefits highlights the limitation of current patient selection approaches. Consequently, and in accordance with the current international guidelines,^{14,18,25,39} EVT should be considered an individualized rescue therapy for selected patients with progressive neurologic deterioration, extensive clot burden, refractory IHS, or radiologic worsening despite optimized anticoagulation, preferably in experienced centers.

DC in CVT

Randomized data are lacking for DC. The strongest prospective evidence comes from DECOMPRESS2,⁴⁰ a multicenter study in which more than one-third of patients achieved a favorable 12-month outcome (modified Rankin Scale [mRS] 0–2) after DC, with generally positive patient and caregiver perspectives. Coma at presentation, preoperative pupillary abnormalities, and advanced age were associated with poor outcomes (mRS 4–6). Current evidence supports considering DC in selected patients with malignant CVT and impending herniation, particularly when neurologic deterioration occurs despite optimal medical therapy.^{40,41} From a clinical standpoint, DC in CVT represents a fundamentally different scenario compared with malignant arterial infarction. Venous infarcts are often hemorrhagic, multifocal, and potentially reversible if venous outflow is restored. Therefore, early recognition of malignant IHS rather than waiting for irreversible brainstem injury may be critical. In our view, surgical decision-making should prioritize neurologic deterioration, radiologic evidence of mass effect and midline shift, and signs of refractory IHS, rather than static imaging findings alone.

Although advanced age and deep coma predict worse outcomes, these factors should not automatically preclude surgical consideration because functional recovery can still occur in carefully selected cases.

Follow-Up and Prognosis

Recanalization and Its Clinical Relevance

Partial or complete venous recanalization occurs in approximately 80% of patients within 3–6 months after CVT diagnosis.^{42,43} Failure to recanalize is associated with worse functional outcomes, persistent headache, and possible higher recurrence risk.⁴³ Early recanalization—sometimes evident within the first week on therapeutic anticoagulation—often parallels regression of nonhemorrhagic parenchymal lesions.⁴² Favorable predictors include early anticoagulation, lower thrombus burden, and absence of malignancy, whereas extensive thrombosis, infection-related CVT, and antiphospholipid syndrome predict lower recanalization rates and may justify prolonged anticoagulation and closer follow-up.^{4,42–44}

Despite good survival, residual symptoms are common, affecting up to two-thirds of survivors, including cognitive complaints, reduced processing speed, fatigue, and mood disorders. Depression and anxiety occur in a significant proportion of patients ($\approx 20\%$ for depression) and require systematic screening and management, given their strong impact on quality-of-life and return-to-work outcomes.⁴⁵

Headache remains the most frequent persistent symptom, reported by 50% during follow-up; $\sim 14\%$ experience severe headaches requiring bed rest or hospitalization.^{9,46} Most post-CVT headaches are new-onset, predominantly tension-type headaches, followed by migraine.⁴⁶ Absent venous recanalization and isolated IHS are risk factors for severe or refractory headache.⁴³ Red flags—new or worsening headache with papilledema, pulsatile tinnitus, focal deficits, or bruit—should prompt repeat venous imaging (MRV/CTV) to exclude dural arteriovenous fistula, recurrent CVT, or IHS.⁴⁶

Neuropsychiatric and Social Outcomes

Although 80%–90% of adults with CVT regain functional independence, residual—often “invisible”—sequelae remain common and clinically relevant.^{45,47} Up to 50% of patients do not return to work within 12 months, and approximately 30% remain unemployed long term, largely because of cognitive impairment, fatigue, mood disorders, and chronic headache rather than motor deficits.^{45,47} Proposed mechanisms of post-CVT cognitive dysfunction may include blood-barrier disruption, neuroinflammation, and possible acceleration of neurodegenerative processes.^{47,48} In addition, failure to recanalize the deep cerebral venous system has been associated with thalamic atrophy and long-term cognitive deficits.⁴⁸

Growing evidence implicates systemic inflammation in both the pathophysiology and prognosis of CVT.^{49–51} Elevated inflammatory mediators, including interleukin-6, interleukin-8, tumor necrosis factor- α , and interferon gamma, have been associated with endothelial dysfunction, thrombogenesis, and hypercoagulability.⁵¹ Experimental models suggest that modulation of neutrophil-mediated inflammation may reduce cerebral edema and blood-barrier disruption. In clinical cohorts, higher levels of inflammatory biomarkers, including neutrophil-to-lymphocyte ratio, C-reactive protein, and interleukin-6 levels, have been associated with worse functional outcomes, although their relationship with lesion progression or venous recanalization remains uncertain.^{50,51}

Emerging evidence suggests that disturbances in cerebral venous outflow may contribute to persistent neurologic sequelae beyond the acute phase of CVT. Altered venous drainage and impaired glymphatic function have been proposed as potential mechanisms underlying cognitive symptoms, fatigue, and other long-term neuropsychological consequences.⁵² Advanced neuroimaging techniques assessing venous hemodynamics, white matter integrity, and glymphatic function may provide further insights into these

processes and help identify novel targets for prognosis assessment and rehabilitation.⁵²

These observations highlight the potential contribution of venous congestion and impaired glymphatic function to long-term neurologic sequelae after CVT. However, current evidence is limited by retrospective designs, small sample sizes, heterogeneous outcome definitions, and inconsistencies in follow-up assessment. Consequently, these findings should be regarded as hypothesis generating until confirmed in large prospective cohorts.

Mortality Scores in CVT

In-hospital mortality ranges from 3% to 15%, with most deaths occurring within the first 2 weeks, typically because of transtentorial herniation resulting from large venous infarctions or intracranial hemorrhage, as well as systemic complications such as pulmonary embolism.^{53,54} Long-term prognosis is generally favorable, with 5-year survival rates exceeding 85% in most cohorts.^{4,44}

Several prognostic models have been developed to predict mortality and poor functional outcome in CVT, most commonly defined as mRS 3–6. The International Study on Cerebral Venous Thrombosis Risk Score incorporates male sex, altered mental status or coma at presentation, deep CVT, and intracerebral hemorrhage to stratify patients according to outcome.⁵⁵ The more recent SI₂NCAL₂C score, proposed by the International CVT Consortium, integrates clinical, laboratory, and etiologic factors—including absence of female-specific risk factors, intracranial hemorrhage, infection, neurologic deficits, coma, age, low hemoglobin, hyperglycemia, and cancer—to predict mortality and functional dependence.⁵⁶ In addition, the CVT-Grading Score classifies patients using lesion size, level of consciousness, hemorrhage, Babinski sign, and sex, providing a highly accurate mortality prediction model.⁵⁷ More recently, the International REvASC⁵⁸ score was developed to predict unfavorable outcomes after CVT and has shown good discrimination in external validation studies, supporting its potential utility across diverse patient populations.

As summarized in Table 2, these models differ substantially in derivation cohorts, outcome definitions, and included predictors. Therefore, they should not be considered interchangeable because each reflects distinct prognostic dimensions of CVT severity and outcome. Current prognostic scores do not incorporate inflammatory markers, venous collateral status, or long-term neurocognitive outcome, all of which may provide additional prognostic information.

From a clinical standpoint, currently available prognostic scores primarily reflect baseline disease severity and may not fully capture the dynamic evolution of CVT. Important factors such as inflammatory activity, venous collateral capacity, thrombus evolution, and long-term neurologic outcomes remain largely absent from existing risk models.

Table 2 Prognostic Scores in Cerebral Venous Thrombosis

Score	Population	Best suited for	Variables included	Strengths	Limitations	Practical point
ISCVT Risk Score⁵⁵	Prospective international cohort	Baseline prognostic, 6-mo functional outcome	• Male gender • Coma • Mental status • Hemorrhage • Malignancy • Deep venous thrombosis	• First validated prognostic score • Widely cited • External validation • Practical • High sensitivity	• Derived from historical cohort • Low specificity	0–9 points Poor outcome ≥3 points
CVT-GS⁵⁷	Single-country tertiary cohort	30-d functional outcome and mortality	• Parenchymal size • Bilateral Babinski sign • Male gender • Hemorrhage • Level of consciousness	• Focused in early outcomes • Practical	• Single country cohort • Long inclusion period • No external validation	0–13 points Mild 0–2 Moderate 3–7 Severe 8–13
SI₂NCAL₂C⁵⁶	International multicenter consortium	6-mo functional outcome, 30-d and 1-y mortality	• Gender • Hemorrhage • Age • Coma • Hemoglobin • Glucose • Malignancy	• Largest real-world data set • Multicenter, international • Strong internal validation	• Complex • Retrospective and prospective data • Long inclusion period • No external validation	Online calculator available
IN-REvASC⁵⁸	International multicenter registry	90-d functional outcome and mortality	• NIHSS • Hemorrhage • Black race • Anemia, substance abuse • Creatinine • Age	• Large cohort, international, multicenter • AUC 0.84	• Retrospective • Loss to follow-up • No external validation	0–35 points Low risk <10 Moderate risk 10–19 High risk ≥20

Abbreviations: AUC = area under the curve; CVT-GS = CVT-Grading Score; IN-REvASC = International REvASC; ISCVT = International Study on Cerebral Venous Thrombosis; NIHSS = NIH Stroke Scale.

Future Directions

Despite substantial advances in the diagnosis and management of CVT, important uncertainties remain regarding disease mechanisms, prognostic stratification, and optimizing treatment strategies. Future research should focus on 3 key areas: understanding pathophysiologic mechanisms, improving diagnostic and prognostic tools, and optimizing therapeutic approaches.

Pathophysiologic Mechanisms

Although venous thrombosis initiates the disease process, the mechanisms underlying tissue injury, neurologic deterioration, and long-term disability remain incompletely understood. Future studies should further characterize the role of thromboinflammation, endothelial dysfunction, blood-brain barrier disruption, and venous hemodynamics in the development of cerebral edema, venous infarction, and hemorrhagic lesions. In addition, the biological basis of divergent clinical trajectories—including why some patients develop malignant intracranial hypertension or extensive parenchymal injury despite similar clot burden—requires further investigation.

Emerging evidence suggests that disturbances in venous drainage and lymphatic function may contribute to persistent cognitive symptoms, fatigue, and long-term neuropsychological sequelae.⁵² Integrating inflammatory biomarkers, such as interleukins and

the neutrophil-to-lymphocyte ratio, with tissue damage biomarkers, such as glial fibrillary acidic protein and neurofilament light, may help identify biologically relevant phenotypes.^{49–51,59} Large prospective multicenter studies incorporating circulating biomarkers will be essential to advance mechanistic understanding and improve risk stratification and probably therapeutic targets.

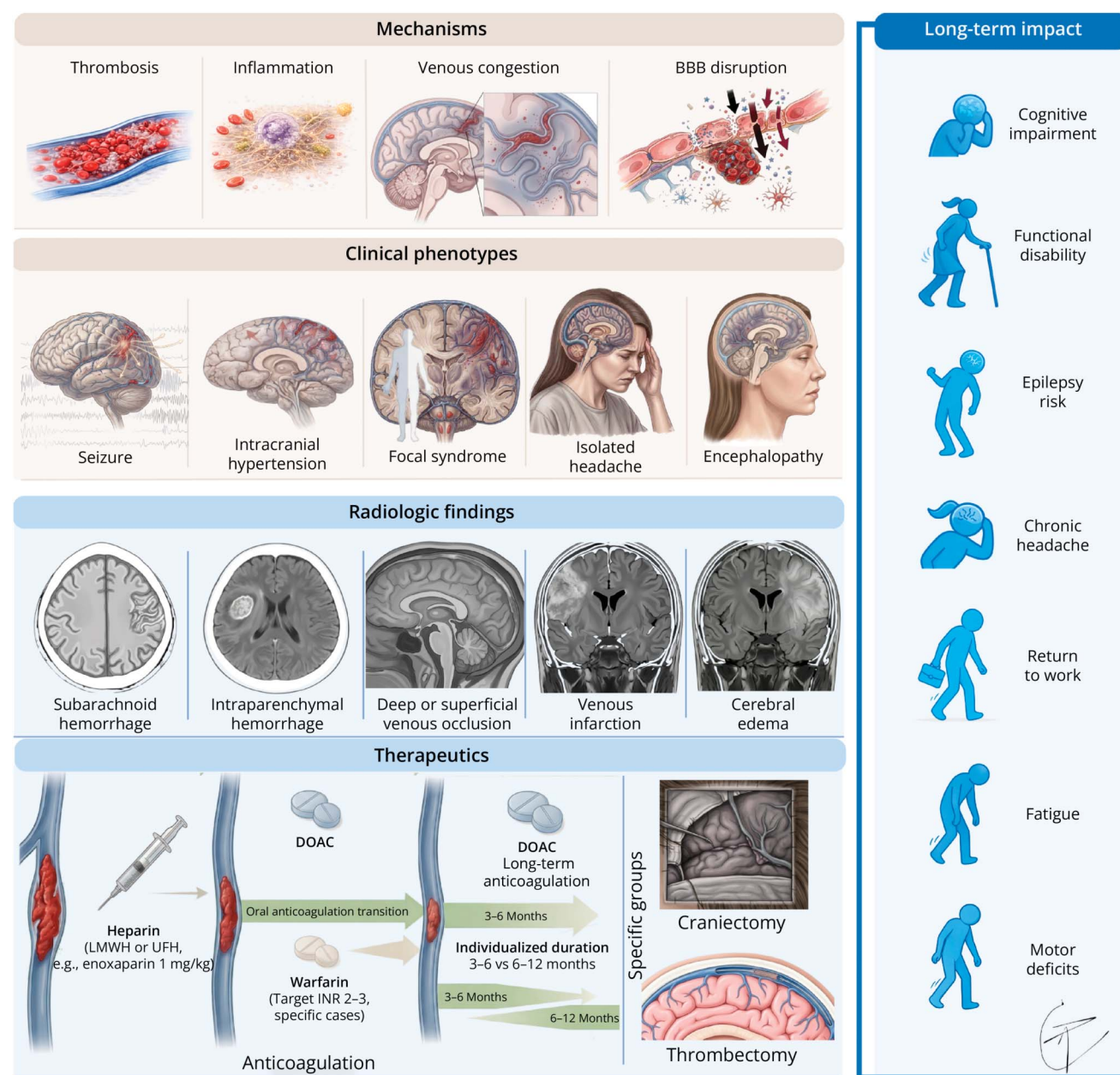
Diagnostic and Prognostic Advances

Improving early diagnosis and prognostic stratification remains a major research priority in CVT. Although MRV and CTV have substantially improved diagnostic accuracy, important uncertainties persist regarding the optimal imaging markers of thrombus evolution, tissue injury, collateral circulation, recanalization and long-term outcomes.

Advanced imaging techniques assessing venous collateral circulation and perfusion patterns may improve risk stratification in CVT. Ongoing studies, such as the PISTE, are evaluating imaging markers of neurologic deterioration that could facilitate earlier identification of patients requiring more aggressive therapeutic approaches.

Current prognostic models have important limitations in capturing the dynamic course of CVT. Artificial intelligence–based approaches integrating clinical and multimodal imaging data may improve both diagnostic accuracy and outcome prediction.⁶⁰

Figure Integrated Overview of CVT



Integrated overview of CVT, illustrating pathophysiologic mechanisms (thrombosis, inflammation, venous congestion, and BBB disruption), clinical phenotypes (seizures, intracranial hypertension, focal deficits, isolated headache, and encephalopathy), radiologic findings (subarachnoid and intraparenchymal hemorrhage, venous occlusion, venous infarction, and cerebral edema), and therapeutic strategies including anticoagulation, DOACs, and selected invasive approaches such as decompressive craniectomy and thrombectomy. Long-term outcomes include cognitive impairment, functional disability, epilepsy risk, chronic headache, fatigue, and motor deficits. BBB = blood-brain barrier; CVT = cerebral venous thrombosis; DOAC = direct oral anticoagulant; LMWH = low-molecular-weight heparin; UFH = unfractionated heparin. © Adrián Pereda Castillo. Used with permission.

Standardized imaging protocols, harmonized recanalization assessment, and externally validated prognostic models will be essential to improve comparability across studies and facilitate the identification of imaging and biological markers associated with neurologic deterioration, treatment response, and long-term outcomes.

Clinical and Therapeutic Studies

Despite advances in anticoagulation and supportive care, several therapeutic uncertainties remain. Key priorities include

defining the optimal duration of anticoagulation, identifying patients who may benefit from extended therapy, and clarifying the role of DOACs in high-risk populations, including patients with thrombophilia, malignancy, provoked and unprovoked thrombosis, recurrence, or persistent recanalization abnormalities.

Although the TO-ACT trial did not demonstrate a clear clinical benefit of EVT,³⁸ important uncertainties remain. Future research should focus on refining patient selection,

optimizing procedural timing, and standardizing endovascular techniques and devices. These efforts may help identify patients most likely to benefit from EVT and better define its role in CVT management.

Finally, future research should extend beyond conventional functional outcomes and focus on chronic headache, cognitive impairment, fatigue, mood disorders, return to work and quality of life. Large multicenter, collaborative studies incorporating standardized outcome definitions and harmonized treatment protocols will be critical to improving long-term care and advancing individualized management strategies in CVT.

Conclusion

CVT remains a complex cerebrovascular disorder in which timely diagnosis and appropriate management are critical for optimal outcomes. Although advances in neuroimaging and anticoagulation have improved survival, persistent cognitive, neuropsychiatric, and functional sequelae remain common, highlighting the need for personalized treatment strategies and long-term rehabilitation. Future progress will depend on refined risk stratification, deeper understanding of disease mechanisms, and the integration of advanced imaging and artificial intelligence–driven tools, with disability defined beyond motor outcomes to include quality of life and return to function (Figure).

Acknowledgment

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Author Contributions

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